Course Project

AI 2024

We will NOT have an exam at the end of the course, but we have a course project for you. We hope that this course project can reflect your scientific research ability or engineering ability, so we designed an optional task like the previous assignment.

Project 1

You need to choose a specific problem in a specific task as your research topic. We can provide some ideas: Long tail distribution problem; Binocular depth estimation problem under extreme weather conditions; Cross-domain generalization problem (this may occur in various tasks), etc.

After determining your topic, you need to complete your report like writing a paper, which means you need to propose a novel solution to your topic and include Introduction, Related Works, Methods and Reference.

Note that we do not require you to experimentally prove the feasibility of the method, so you do not need to include an Experiment section, nor do you need to perform actual experiments. However, you still need to explain your method in detail in the Method chapter, which means that although your method may not be feasible in practice, it must be at least feasible in theory.

If you choose this project, you need to:

(1) Research the latest work
(2) Summarize these works and analyze them
(3) Propose your own improvements to the shortcomings of current methods
(4) Write an essay based on the above requirements with at least 6 pages

Here are some tips:

(1) We recommend you use Latex to write the essay
(2) We recommend you use CVPR or Elsevier templates

Project 2

You need to implement code that accomplishes some of the following tasks:

(1) Object Detection: This means that you need to output the bounding box and classification results, and perform visual display and experiment on test set. The dataset you use MUST be chosen from the following list:

- PASCAL VOC 2007
- MSCOCO 2014

(2) Facial Expression Recognition: This is a relatively simple task that usually only requires classification based on the input image, which means it is actually an image classification task. You need to perform visual display and experiment on test set. The

dataset you use MUST be chosen from the following list:

- FER2013
- AffectNet
- EmoReact
- RAF-DB (Real-world Affective Faces Database)
- SFEW（Static Facial Expressions in the Wild）

(3) Semantic Segmentation: As we discussed in class, you need to generate corresponding classification labels for each pixel. You need to perform visual display and experiment on test set. The dataset you use MUST be chosen from the following list:

- ADE20K
- cityScapes
- VOC12_AUG

If you choose this project, you need to:

(1) Programming in any programming language by oneself
(2) Clarify the input and output forms of the task
(3) Analyze data
(4) Analyze the effectiveness of the algorithm and visualization

Here are some tips:

(1) You can refer to open-source code, but do not plagiarize
(2) There is not much textual content, but it is still necessary to explain the experimental results. The "Experiment" section in this paper is similar. This section should include an introduction to experiment, the dataset and an analysis of the experimental results

Submission:

Deadline: 12/31/2024

Format: **Project 1:** An essay in PDF format

Naming: Student Number_Name_Project1.pdf

Project 2: An compressed file with your source code and a analysis report in PDF format

Naming: Student Number_Name_Project2_Code.zip/rar

Student Number_Name_Project2_Report.pdf

Submit by sending to submittal_bai@163.com